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FURTHER MATHEMATICS

9231/31

Paper 3 Further Mechanics

May/June 2023

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

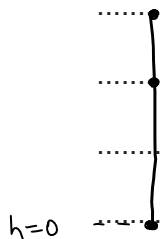
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- 1 One end of a light elastic string, of natural length a and modulus of elasticity $3mg$, is attached to a fixed point O . The other end of the string is attached to a particle P of mass m . The string hangs with P vertically below O . The particle P is pulled vertically downwards so that the extension of the string is $2a$. The particle P is then released from rest.

- (a) Find the speed of P when it is at a distance $\frac{3}{4}a$ below O . [3]

Initially:



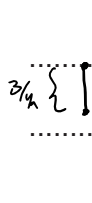
$$PE = 0$$

$$KE = 0$$

$$EPE = \frac{\lambda x^2}{2l} = \frac{3mg(2a)^2}{2a} = 6mga \text{ J}$$

Finally:

→ By CoE, energy of system = $6mga$.



$$\left. \begin{array}{l} EPE = 0 \\ PE = \frac{3}{4}mga \end{array} \right\} \begin{array}{l} KE = \left(6 - \frac{3}{4}\right)mga \\ = \frac{15}{4}mga = \frac{1}{2}m\left(\sqrt{\frac{15}{2}ga}\right)^2 \end{array} \rightarrow \therefore v = \sqrt{7.5ga}$$

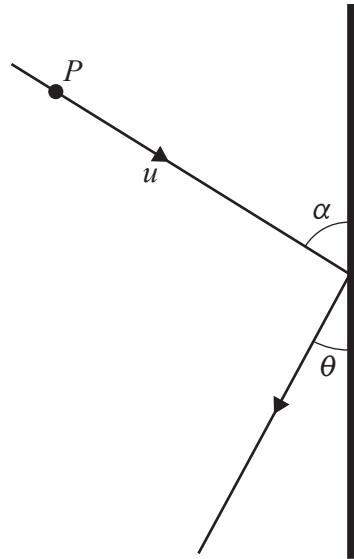
- (b) Find the initial acceleration of P when it is released from rest. [2]

$$T = \frac{\lambda x}{l} = \frac{3mg(2a)}{a} = 6mg$$

$$W = mg$$

net force = $5mg$ upward.

$$a = 5g \text{ ms}^{-2}$$



A particle P of mass m is moving with speed u on a fixed smooth horizontal surface. It collides at an angle α with a fixed smooth vertical barrier. After the collision, P moves at an angle θ with the barrier, where $\tan \theta = \frac{1}{2}$ (see diagram). The coefficient of restitution between P and the barrier is e . The particle P loses 20% of its kinetic energy as a result of the collision.

Find the value of e .

[5]

Only horizontal components are affected by this collision



$$\text{Final KE} = 0.8 \left(\frac{1}{2} m u^2 \right)$$

$$= \frac{1}{2} m \left(\frac{2\sqrt{5}}{5} u \right)^2 \Rightarrow \boxed{\text{final } v = \frac{2\sqrt{5}}{5} u}$$

$$v_y = v_y = \frac{2\sqrt{5}}{5} u \cos \theta = \frac{4}{5} u$$

$$\text{we know that } \sqrt{v_x^2 + v_y^2} = v$$

$$\hookrightarrow v^2 = v_x^2 + \frac{16}{25} u^2$$

$$\frac{9}{25} u^2 = v_x^2$$

$$v_x = \frac{3}{5} u$$

$$\sqrt{v_x^2 + v_y^2} = v$$

$$v_x^2 + \frac{16}{25} u^2 = \left(\frac{2\sqrt{5}}{5} u \right)^2$$

$$v_x^2 = \frac{4}{25} u^2$$

$$v_x = \frac{2}{5} u$$

$$v_x = e v_x$$

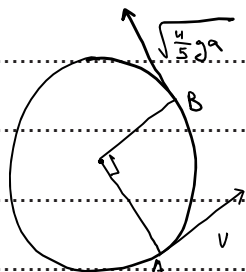
$$e = \frac{\frac{2}{5} u}{\frac{3}{5} u} = \boxed{\frac{2}{3}}$$

- 3 A particle P of mass m is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point O . The particle P is held at the point A , where OA makes an angle θ with the downward vertical through O , and with the string taut. The particle P is projected perpendicular to OA in an upwards direction with speed u . It then starts to move along a circular path in a vertical plane. The string goes slack when P is at B , where angle AOB is 90° and the speed of P is $\sqrt{\frac{4}{5}ag}$.

- (a) Find the value of $\sin \theta$.

180- (0+90)

[2]



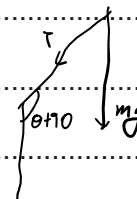
Considering forces @ B:

$$T = \frac{mv^2}{r} + mg \cos(\theta + 90)$$

$$\frac{mv^2}{r} = mg \sin \theta$$

$$\frac{4}{5}ag = g \sin \theta$$

$$\sin \theta = \frac{4}{5}$$



- (b) Find, in terms of m and g , the tension in the string when P is at A .

[5]

Taking speed at bottom of circle = v

$$T_A = \frac{mv^2}{a} - 2mg + 3mg \cos \theta \Rightarrow \cos \theta = \frac{3}{5}$$

$$= \frac{mv^2}{a} - \frac{mg}{5}$$

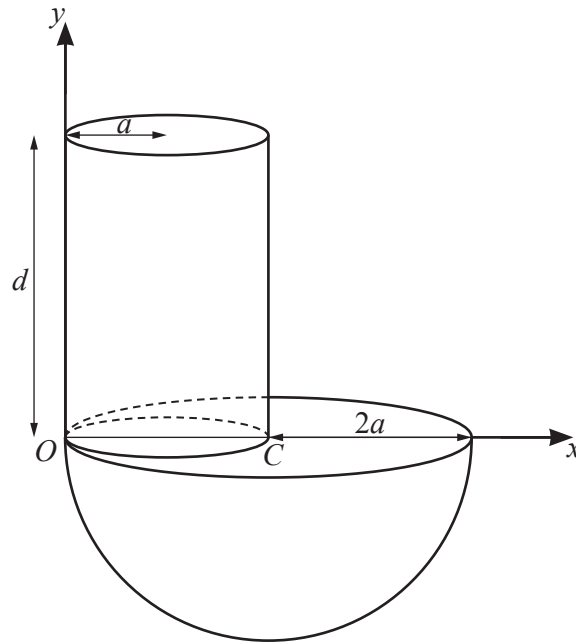
$$0 = T_B = \frac{mv^2}{a} - 2mg + 3mg \cos(\theta + 90)$$

$$\frac{mv^2}{a} = 2mg + \frac{12}{5}mg$$

$$v^2 = 4.4ga$$

$$T_A = \frac{m(4.4ga)}{a} - \frac{mg}{5}$$

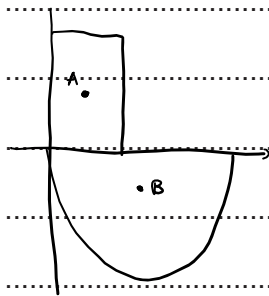
$$= 4.2mg$$



An object is formed from a solid hemisphere, of radius $2a$, and a solid cylinder, of radius a and height d . The hemisphere and the cylinder are made of the same material. The cylinder is attached to the plane face of the hemisphere. The line OC forms a diameter of the base of the cylinder, where C is the centre of the plane face of the hemisphere and O is common to both circumferences (see diagram). Relative to axes through O , parallel and perpendicular to OC as shown, the centre of mass of the object is $(\bar{x}, \bar{y})$.

- (a) Show that $\bar{x} = \frac{32a^2 + 3ad}{16a + 3d}$ and find an expression, in terms of a and d , for $\bar{y}$. [5]

By symmetry, $(\bar{x}, \bar{y})$ lies on the axis, so we need only consider a 2D image:



$$A: (a, \frac{d}{2})$$

$$B: (2a, -\frac{3a}{4})$$

About x :

$$(\bar{x})\left(\frac{2}{3}\pi(2a)^3\rho + \pi a^2 d\rho\right) = \left(\frac{2}{3}\pi(2a)^3\rho\right)(2a) + (\pi a^2 d\rho)(a)$$

$$(\bar{x})(\pi a^2\rho)\left(\frac{16}{3}a + d\right) = (\pi a^2\rho)\left(\frac{32}{3}a^2\right) + (\pi a^2\rho)(ad)$$

$$\bar{x} = \frac{\frac{32}{3}a^2 + ad}{\frac{16}{3}a + d} = \frac{32a^2 + 3ad}{16a + 3d}$$

About y :

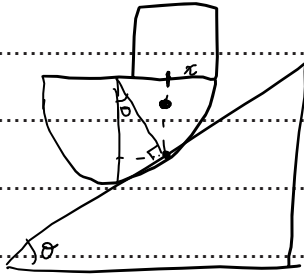
$$\bar{y}(\pi a^2\rho)\left(\frac{16}{3}a + d\right) = (\pi a^2\rho)(-4a^2) + (\pi a^2\rho)\left(\frac{d^2}{2}\right)$$

$$\bar{y} = \frac{\frac{d^2}{2} - 4a^2}{\frac{16}{3}a + d} = \frac{3d^2 - 24a^2}{32a + 6d}$$

The object is placed on a rough plane which is inclined to the horizontal at an angle θ where $\sin \theta = \frac{1}{6}$. The object is in equilibrium with CO horizontal, where CO lies in a vertical plane through a line of greatest slope.

(b) Find d in terms of a .

[3]



$$\text{COR: } \left(\frac{32a^2 + 3ad}{16a + 3d}, \frac{3d^2 - 24a^2}{32a + 6d} \right)$$

$$\rightarrow \text{We need to find } z, \text{ as } z = \frac{32a^2 + 3ad}{16a + 3d}$$

$$\sin \theta = \frac{2a - z}{2a}$$

$$z = 2a - 2a \sin \theta$$

$$= \frac{5a}{3}$$

$$\hookrightarrow \frac{32a^2 + 3ad}{16a + 3d} = \frac{5a}{3}$$

$$96a^2 + 9ad = 80a^2 + 15ad$$

$$16a + 9d = 80a + 15d \quad \text{as } a \neq 0$$

$$16a = 6d$$

$$d = \frac{8a}{3}$$

- 5 A light elastic string of natural length a and modulus of elasticity λmg has one end attached to a fixed point O on a smooth horizontal surface. When a particle of mass m is attached to the free end of the string, it moves with speed v in a horizontal circle with centre O and radius x . When, instead, a particle of mass $2m$ is attached to the free end of the string, this particle moves with speed $\frac{1}{2}v$ in a horizontal circle with centre O and radius $\frac{3}{4}x$.

(a) Find x in terms of a .

[5]

In both cases, Tension = Centripetal force

$$\textcircled{1} \quad \frac{(\lambda mg)(x-a)}{a} = \frac{mv^2}{x}$$

$$\lambda gx^2 - \lambda gax = av^2$$

$$\textcircled{2} \quad \frac{(\lambda mg)(\frac{3}{4}x-a)}{a} = \frac{2m(\frac{1}{2}v)^2}{\frac{3}{4}x}$$

$$\frac{9}{16} \lambda gx^2 - \frac{3}{4} \lambda gxa = \frac{1}{2} av^2$$

Equating:

$$\frac{9}{8} \lambda gx^2 - \frac{3}{2} \lambda gxa = \lambda gx^2 - \lambda gxa$$

$$\frac{9}{8}x - \frac{3}{2}a = x - a$$

$$\frac{1}{8}x = \frac{1}{2}a$$

$$\boxed{x = 4a}$$

(b) Given that $v = \sqrt{12ag}$, find the value of λ .

[2]

$$\lambda g x^2 - \lambda g x a = a v^2$$

$$\hookrightarrow \lambda g (4a)^2 - \lambda g (4a)(a) = a(12ag)$$

$$16\lambda g a^2 - 4\lambda g a^2 = 12g a^2$$

$$12\lambda = 12$$

$$\boxed{\lambda = 1}$$

- 6 A particle P moving in a straight line has displacement x m from a fixed point O on the line and velocity v m s⁻¹ at time t s. The acceleration of P , in m s⁻², is given by $6v\sqrt{v+9}$. When $t = 0$, $x = 2$ and $v = 72$.

(a) Find an expression for v in terms of x .

[4]

$$a = 6v\sqrt{v+9}$$

We know:

$$\frac{dx}{dt} \left(\frac{dv}{dx} \right) = \frac{dv}{dt}$$

$$\frac{dv}{dx} = \frac{a}{v}$$

∴

$$\frac{dv}{dx} = 6\sqrt{v+9}$$

$$\hookrightarrow \frac{1}{\sqrt{v+9}} dv = 6 dx$$

$$\int (v+9)^{-1/2} dv = \int 6 dx$$

$$2(v+9)^{1/2} = 6x + c$$

$$\sqrt{v+9} = 3x + c$$

∴ when $x=2$, $v=72$, so:

$$9 = 6 + c$$

$$\hookrightarrow c = 3$$

$$v+9 = (3x+3)^2$$

$$v = 9x^2 + 18x$$

(b) Find an expression for x in terms of t .

[5]

$$\frac{dx}{dt} = 9x^2 + 18x$$

$$\frac{1}{x^2 + 2x} dx = 9 dt$$

$$\int \frac{1}{x^2 + 2x} dx = \int 9 dt$$

$$\hookrightarrow \frac{1}{x^2 + 2x} = \frac{A}{x+2} + \frac{B}{x}$$

$$\hookrightarrow Ax + Bx + 2B = 1$$

$$\hookrightarrow B = \frac{1}{2}$$

$$A = -\frac{1}{2}$$

$$\Rightarrow \int \frac{1}{2x} - \frac{1}{2x+4} dx = 9t + C$$

$$\int \frac{1}{x} - \frac{1}{x+2} dx = 18t + C$$

$$\ln|x| - \ln|x+2| = 18t + C$$

$$\hookrightarrow \text{when } t=0, x=2$$

$$\hookrightarrow \ln \frac{2}{4} = C$$

$$C = \ln \frac{1}{2}$$

$$\ln \left| \frac{x}{x+2} \right| = 18t + \ln \frac{1}{2}$$

$$\ln \left| \frac{2x}{x+2} \right| = 18t$$

$$\frac{2x}{x+2} = e^{18t}$$

$$2x = e^{18t} x + 2e^{18t}$$

$$x = \frac{2e^{18t}}{2 - e^{18t}}$$

- 7 At time t s, a particle P is projected with speed 40 ms^{-1} at an angle θ above the horizontal from a point O on a horizontal plane and moves freely under gravity. The greatest height achieved by P during its flight is H m and the corresponding time is T s.

(a) Obtain expressions for H and T in terms of θ .

[2]

$$v_y = u \sin \theta - gt \Rightarrow \text{@ max height, } v_y = 0, \text{ so } t = \frac{u \sin \theta}{g} = \boxed{4 \sin \theta}$$

$$\begin{aligned} H &= u \sin \theta T + \frac{1}{2}(-g)(T)^2 = (40 \sin \theta)(4 \sin \theta) - \frac{1}{2}(10)(4 \sin \theta)^2 \\ &= 160 \sin^2 \theta - 80 \sin^2 \theta \\ &= \boxed{80 \sin^2 \theta} \end{aligned}$$

During the time between $t = T$ and $t = 3$, P descends a distance $\frac{1}{4}H$.

(b) Find the value of θ .

[4]

$$\begin{aligned} \text{@ } t = T, \quad H &= 80 \sin^2 \theta \\ \text{@ } t = 3, \quad H &= 60 \sin^2 \theta \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} 20 \sin^2 \theta \text{ m descended in} \\ (3 - T) \text{ s} \end{array}$$

$$\begin{aligned} \hookrightarrow s &= 20 \sin^2 \theta \\ t &= 3 - 4 \sin \theta \end{aligned} \quad \left(\begin{array}{l} \text{taking down as +} \end{array} \right)$$

$$\begin{aligned} a &= g \\ u &= 0 \end{aligned}$$

$$20 \sin^2 \theta = \frac{1}{2}(g)(3 - 4 \sin \theta)^2$$

$$4 \sin^2 \theta = 9 + 16 \sin^2 \theta - 24 \sin \theta$$

$$0 = 12 \sin^2 \theta - 24 \sin \theta + 9$$

$$= 4 \sin^2 \theta - 8 \sin \theta + 3$$

$$(2 \sin \theta - 3)(2 \sin \theta - 1)$$

$$\hookrightarrow \sin \theta = \frac{1}{2}$$

$$\hookrightarrow \theta = 30^\circ$$

(c) Find the speed of P when $t = 3$.

[3]

$$\begin{aligned}v_y &= 40 \sin \theta - g(3) \\&= 40 \left(\frac{1}{2}\right) - 30 \\&= -10 \text{ ms}^{-1}\end{aligned}$$

$$v_x = v \cos \theta = 40 \left(\frac{\sqrt{3}}{2}\right) = 20\sqrt{3} \text{ ms}^{-1}$$

$$\text{speed} = \sqrt{100 + 1200} = 10\sqrt{13} \text{ ms}^{-1}$$

[illegible]

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